

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMODYNAMICS

LECTURE-02 **PHYSICS**

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Topics To Be Covered

1 Work / heat / internal energy / 1st law of thermodynamics

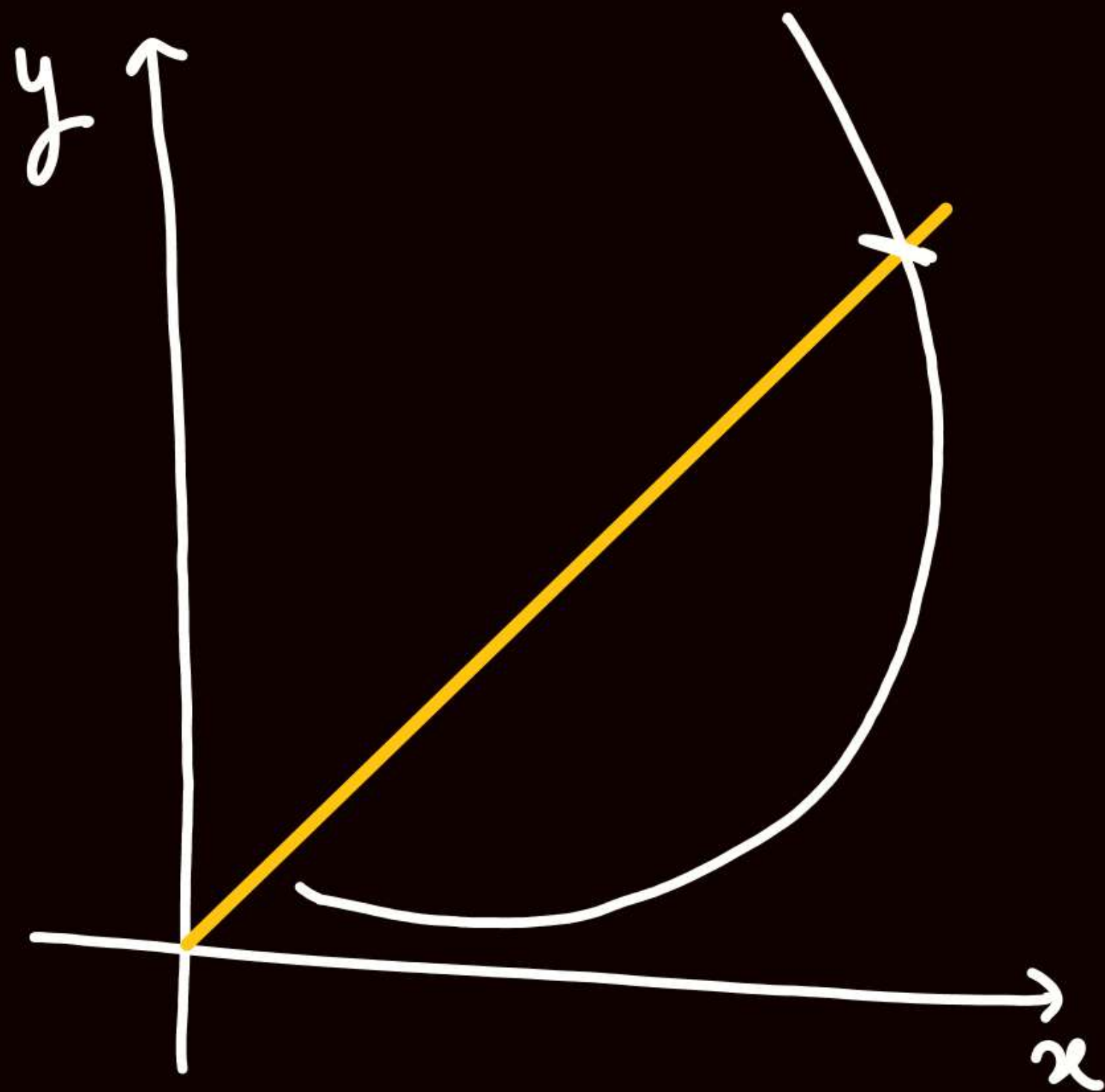
2 ..

Basics $\Rightarrow$ Top

3 ..

4 ..

5 ..



Recap - ①

• Thermodynamics

- System
- Surrounding
- Universe

- Intensive property
- Extensive property
- Isothermal
- Isobaric
- Isochoric
- Adiabatic

- Reversible
- Irreversible
- Cyclic process
- Temperature
- Thermal equilibrium
- Zeroth law of thermodynamics

Thermodynamics

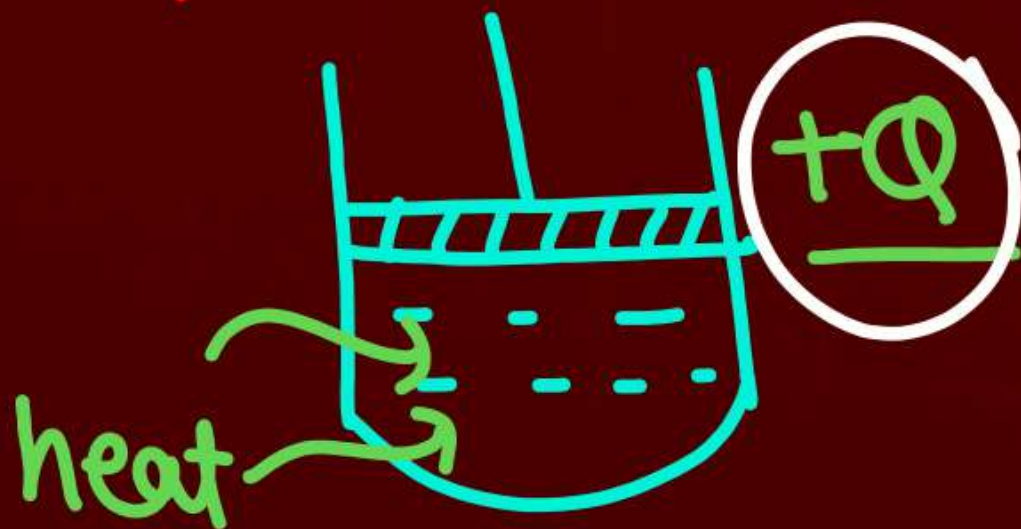
Thermal equilibrium and definition of temperature, zeroth law of thermodynamics, heat, work and internal energy. First law of thermodynamics, Second law of thermodynamics: Gaseous state of matter. change of condition of gaseous state-isothermal, adiabatic, reversible, irreversible, and cyclic processes.

⇒ The form of Energy which produce the Sensation of warmth ①

⇒ Q_p

⇒ Heat unit = Joule

$$Q_p = n C_p \Delta T$$

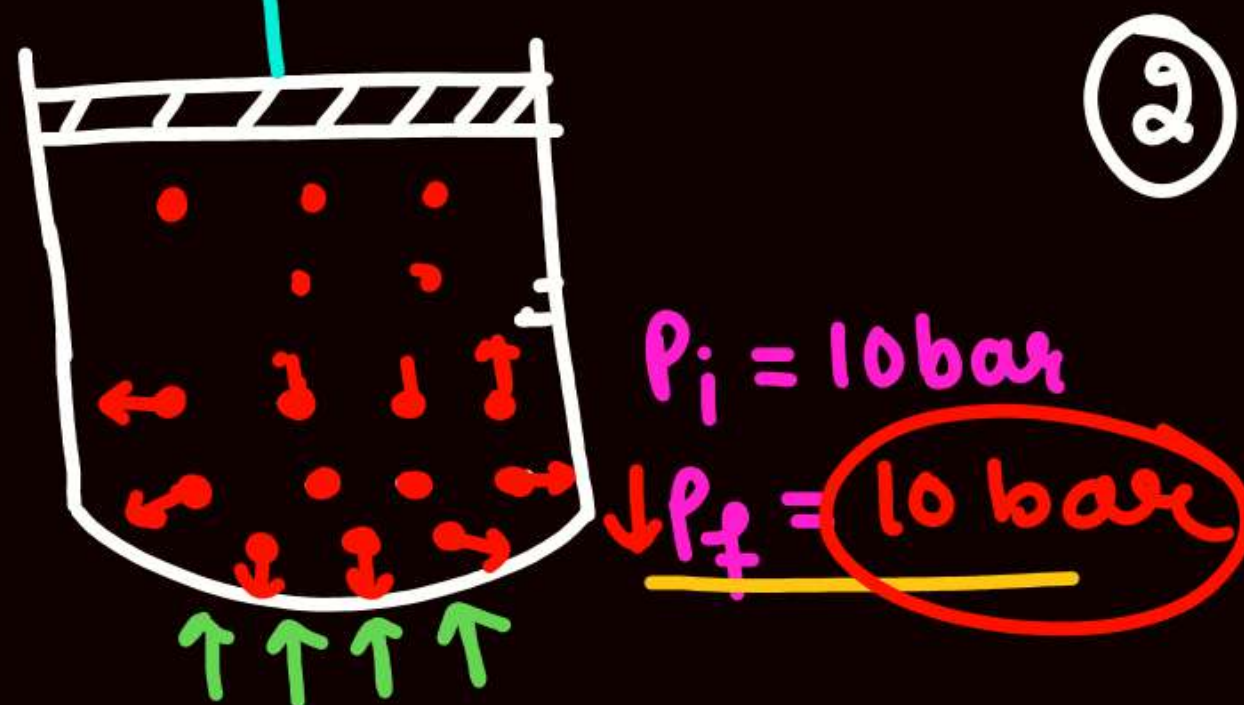
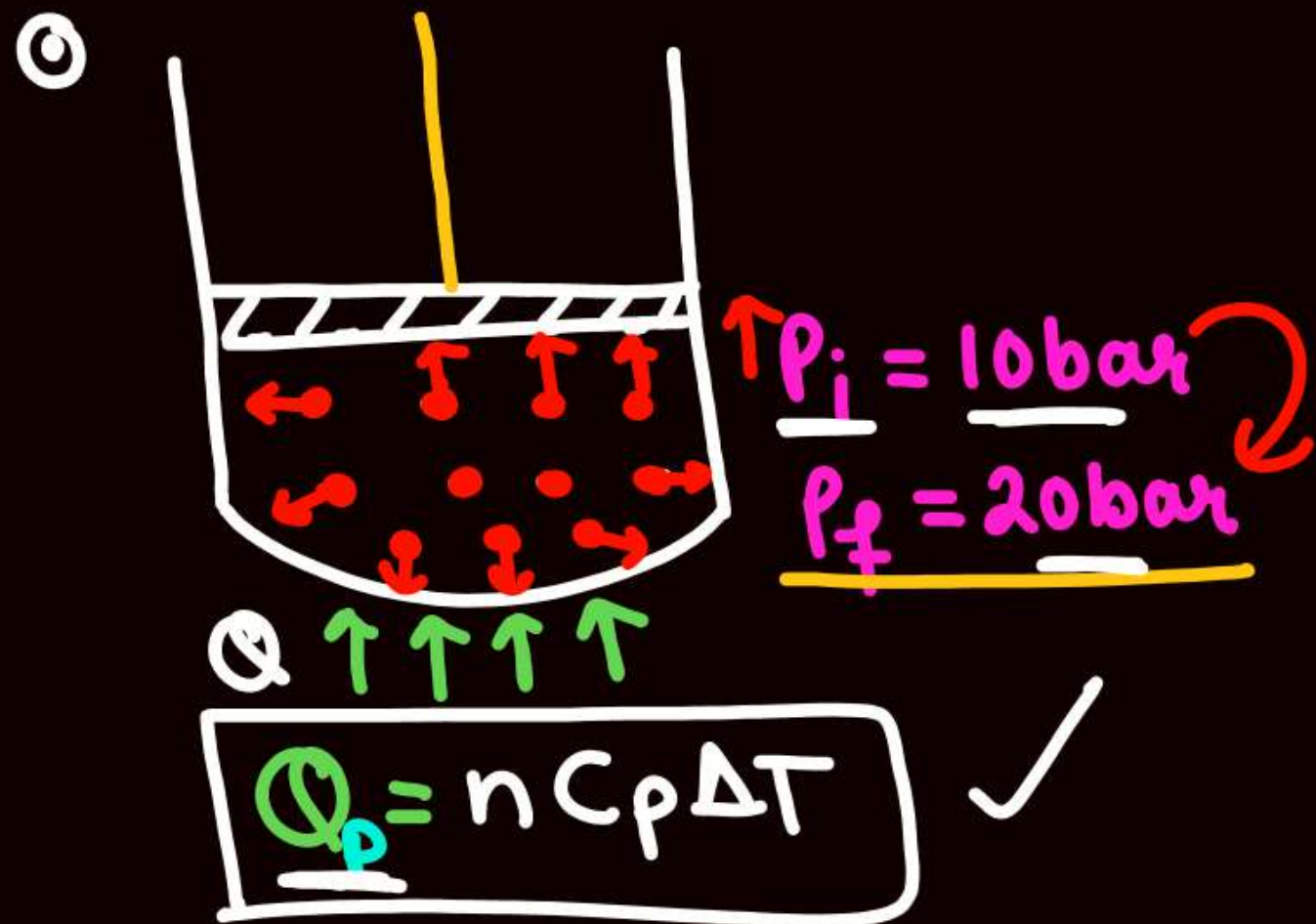


#



$Q = -ive$





(Isobaric)
 Process)

Pressure = constant

$$P_i = P_f$$

$$0 = P_f - P_i$$

$$0 = \Delta P$$

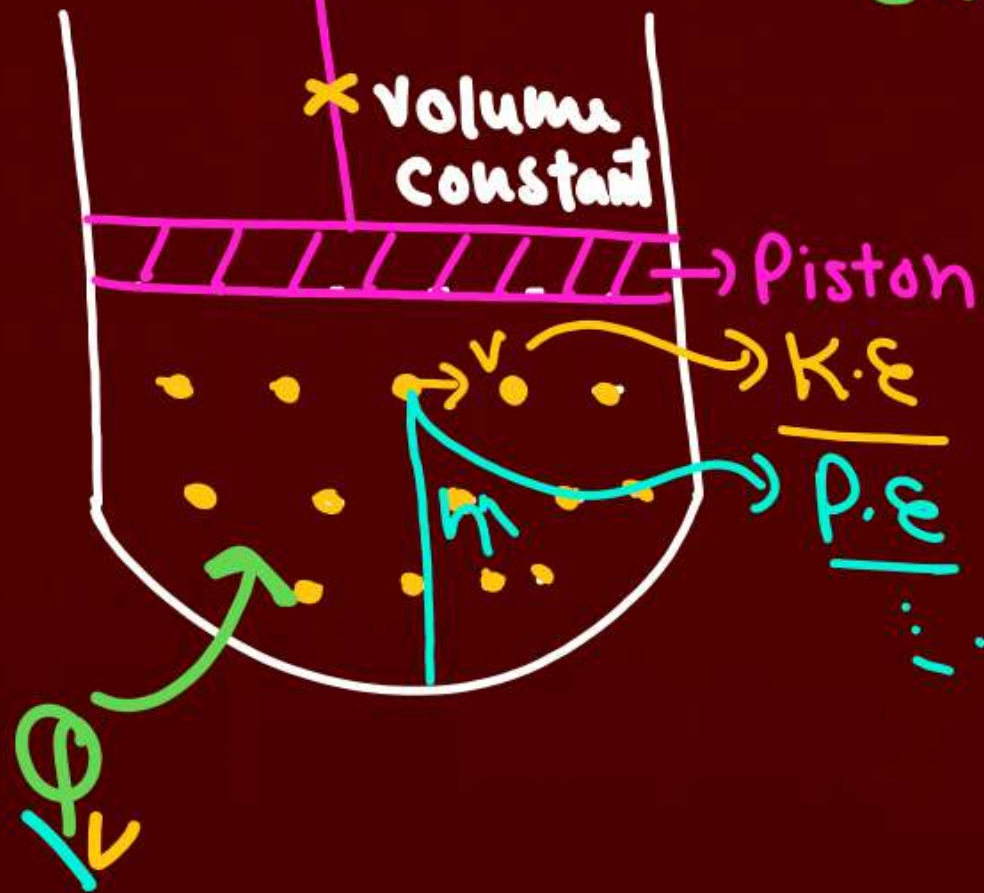
Internal Energy (U)

unit = Joule $\rightarrow$ 2023



- The Sum total of all the Energy of the **gas molecule**

$\Rightarrow$ Piston do not move above



Add Q $U = +ive \uparrow \uparrow \uparrow$
Subtract Q $U = -ive \downarrow \downarrow \downarrow$

$$Q_v = n C_v \Delta T$$

$$U = n C_v \Delta T$$



Work Done

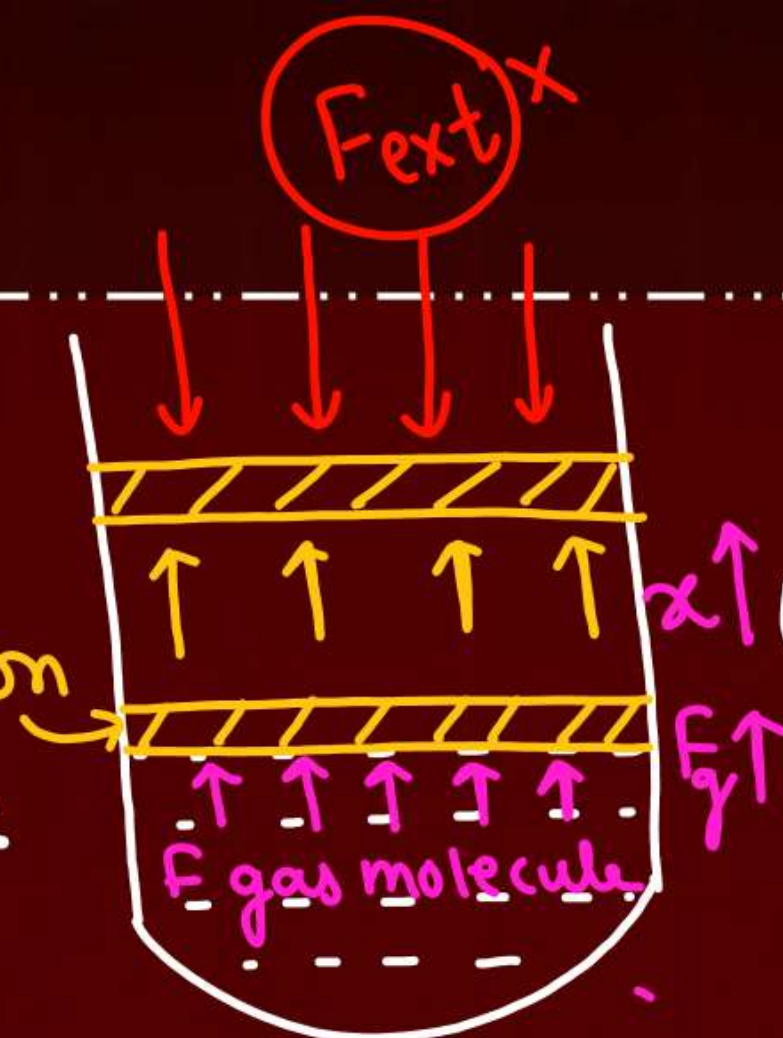
⇒ Thermodynamic work

⇒ Work is the mode of transfer of Energy

$$\Rightarrow W = \vec{F} \cdot \vec{x}$$
$$W = Fx \cos \theta$$

⇒ Scalar

⇒ Joule



⇒ Expansion

WORK DONE By Gas. 4

$$\theta = 0 \quad W = Fx \cos \theta$$

$$W = Fx \cos 0$$

$$W = Fx \times 1$$

$$W = +Fx$$

gas molecule energy loose

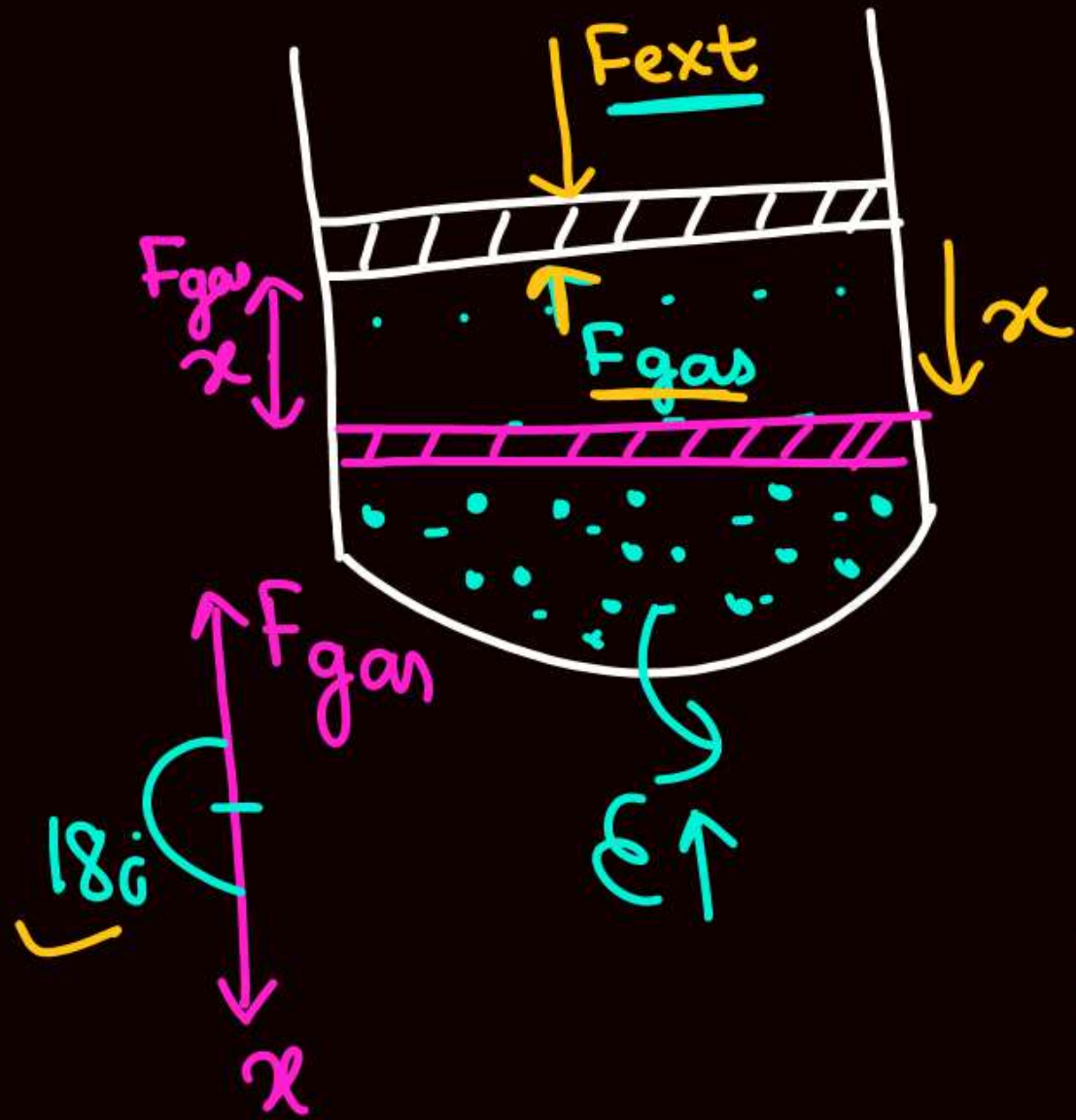
MCQ
2023
2022
2019



Compression

WORK DONE ON GAS.

⑤



$$W = Fx \cos \theta$$

$$W = Fx \cos 180^\circ$$

$$W = Fx \times -1$$

$$W = -Fx$$

(mcQ)

2021

2022

gas molecule
Energy $\uparrow \uparrow \uparrow$

Another formula of Thermodynamic work

(i) Constant pressure

⇒ Isobaric process

$$W = Fx \cos \theta$$

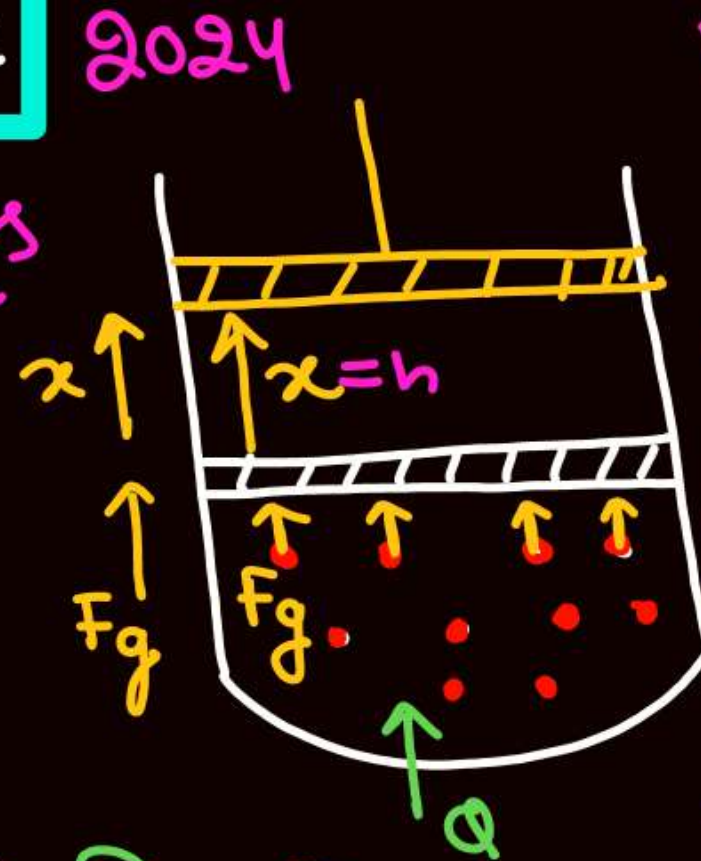
$$W = Fx \cos 0^\circ$$

$$W = Fx \times 1$$

$$W = +fx$$

$$\Rightarrow P = \frac{F}{A}$$

$$PA = F$$



$$V = l \times b \times h$$

$$\underline{V} = \frac{A}{A \times x}$$

$$P_i = 10 \text{ bar}$$

$$P_f = 20 \text{ bar}$$

$$\underline{P_f} = 10 \text{ bar}$$

$$W = Fx \cos \theta$$

$$W = \underline{F} \underline{x} \xrightarrow{PA} \text{ (6)}$$

$$W = P \underline{A} \underline{x}^V$$

$$\boxed{W = +PV} \text{ (1)}$$

Expansion
 $PV \rightarrow \text{WORK}$

$$\boxed{W = -PV}$$

Compression.

Note

→ Expansion

$$W = +P \Delta V$$

$$W = +P \Delta V$$

⑦

→ Compression

$$W = -P \Delta V$$

$$W = -P \Delta V$$

Q what is the unit of PV

(a) Newton

~~(b) Joule~~

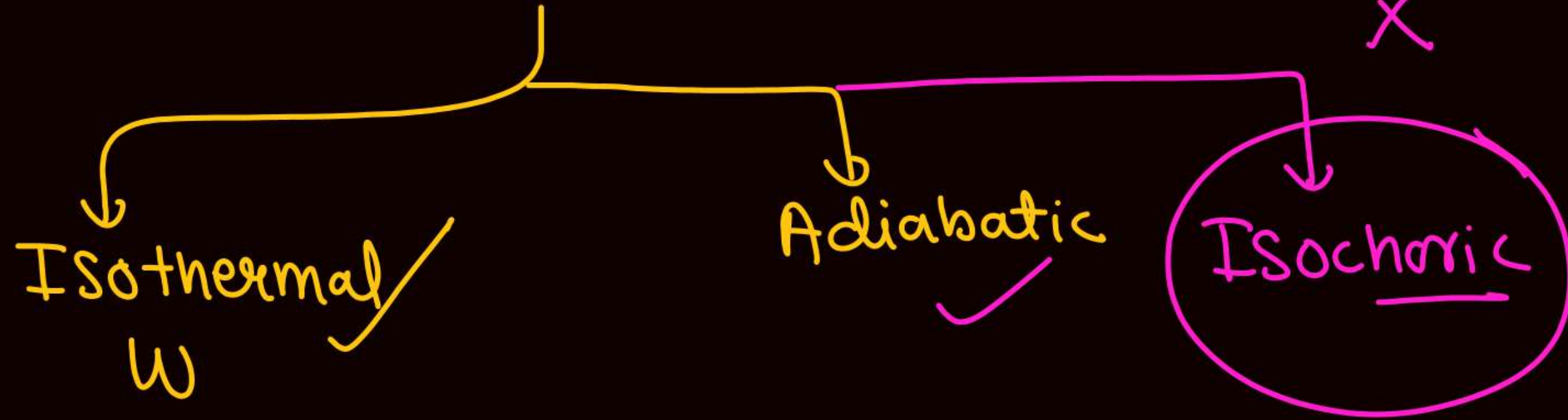
(c) Pascal

(d) main gada hu.

$$\begin{array}{c} \textcircled{W} = \textcircled{PV} \\ \downarrow \quad \downarrow \\ \text{Joule} \quad \text{Joule} \end{array}$$

8

WORK $\rightarrow$ ABBII Bachta hai



Ideal Gas Equation

- The ideal gas equation is the Relationship b/w Pressure^(P), volume^(V), no. of moles⁽ⁿ⁾, and Temperature^(T) of gas molecule

$$\# \boxed{PV = nRT}$$

$$\underline{P\Delta V = nR\Delta T}$$

Derive

$$\bullet \uparrow P \propto \frac{1}{V \downarrow} \text{--- (1)}$$



$$\bullet \uparrow P \propto T \uparrow \text{--- (2)}$$



$$\bullet P \propto n \text{--- (3)}$$

on combining (3) equation

$$P \propto \frac{Tn}{V}$$

$$P \propto \frac{nT}{V}$$



$$P \propto \frac{nT}{V}$$

$$PV \propto nT$$

$$PV = RnT$$

universal
gas constant

$$PV = nRT$$

$R \Rightarrow$ universal gas constant

$$R = 8.14 \frac{\text{Joule}}{\text{mol K}}$$

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unit of R

(a) $\frac{N}{\text{mole K}}$

(b) $\frac{\text{Joule}}{\text{mol K}}$

(c) Joule mole

(d) $\frac{\text{mole}}{\text{Joule K}}$

unit of R

11

$$PV = nRT$$

Joule = mol R K

$$\frac{\text{Joule}}{\text{mol K}} = R$$

note

$$* PV = nRT$$

$$P \Delta V = nR \Delta T$$

⑫

1st Law of thermodynamic

- It is based on the law of conservation of energy.
- Heat supply least increase the internal energy and the does work done.



Relationship between C_p and C_v



Heat Capacity Ratio (γ)



Work Done By Isothermal Process



SUMMARY



$$\begin{array}{l} Q_p \begin{array}{l} \nearrow +Q \\ \searrow -Q \end{array} \\ Q_v / U \begin{array}{l} \nearrow +U \\ \searrow -U \end{array} \end{array}$$

$$\text{WORK} \begin{array}{l} \nearrow W+ \\ \searrow W- \end{array}$$

$$\begin{aligned} \textcircled{1} W &= P \Delta V \\ &= P \Delta V \end{aligned}$$

$$\Rightarrow P \Delta V = n R \Delta T \checkmark$$

HOMework

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2017



Q what is the C.G.S unit $\textcircled{R} = ?$

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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You

